

Dhruv Thippasandra

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Education

University of Southern California

Aug 2025 – May 2027 (On track)

Master of Science in [Electrical Engineering \(Data Science and Machine Learning\)](#) 

- Relevant Coursework: Probability for ECE, Computational Deep Learning, ML: Supervised Methods, Database Systems

University of California, Riverside

Sep 2022 – Jun 2025

Bachelor of Science in [Electrical Engineering](#) 

- GPA: 3.77/4.0
- Relevant Coursework: Machine Learning & Data Mining, Digital Signal Processing, Probability, Random Variables & Random Processes in ECE, Electronic Circuits I & II, Signals and Systems I & II, Embedded Systems, Automatic Control.
- Honors: Graduated **Cum Laude**; **Chancellor's Honors List** (all years); **Dean's Honors List** (all quarters)

Work Experience

Software Engineering Intern

Jun 2026 – Aug 2026

AT&T

- Engineered a scalable backend anomaly detection service using **asynchronous FastAPI endpoints**, integrating an **Extended Isolation Forest (EIF)** model to automatically detect and suspend telecom fraud.
- Built automated **Azure ETL pipelines** to ingest **12M+ phone records** from Snowflake to PostgreSQL, executing bi-hourly incremental upserts of user telemetry metrics to trigger batch fraud inference.
- Deployed a continuous production **MLOps pipeline** retraining models on **5M+** monthly records, achieving high-precision anomaly scoring with an estimated **<0.5% false-positive rate** to safely trigger automated suspensions.
- Streamlined fraud operations** by implementing a dashboard, enabling cross-functional teams to monitor automated enforcement actions, view aggregated statistics about banned users, and maintain platform trust.

Machine Learning Engineer

Sep 2025 – Jan 2026

Mercor

- Compiled external ML competitions into challenging tasks reflecting real-world ML responsibilities. Drafted detailed natural-language execution plans and implemented them in Python within a Docker environment, validating implementations against original plans and marking discrepancies. These tasks are used by researchers to train models on high-quality data.

Projects

Network Degradation Prediction Using Real Internet Telemetry

2025 – 2026 School year

- Engineered a predictive DNS telemetry pipeline with XGBoost on **327MB** of real RIPE Atlas telemetry (Root DNS anchor) to predict network degradation **5 minutes** ahead, achieving **91% accuracy** and **40% recall**.
- Deployed on **AWS EC2** with Docker and **CI/CD** via GitHub Actions; decoupled inference from serving via cron-based background refresh, achieving **sub-100ms** load time with zero model cold-start latency.

Wildfire Spread Forecaster Hosted on Proxmox Home-Cloud

2025 – 2026 School year

- Architected a 3-node bare-metal **Proxmox GPU cluster** orchestrated via **Nomad**, saving **~\$400/mo** in cloud infrastructure costs and serving live microservices ([dhruvq.com](#) , [fire.dhruvq.com](#) ) with zero open ports via Cloudflare Tunnels.
- Optimized geospatial pipeline throughput by implementing concurrent spatial data pre-fetching via Python **asyncio.gather** and in-memory caching, eliminating sequential API bottlenecks across USGS and NOAA endpoints.
- Engineered an async **FastAPI backend** on the cluster fusing live NASA, NOAA, and USGS geospatial data, computing a 6-hour cellular automata fire spread forecast in **<10 seconds** at **75.4% IoU accuracy**.

Apollo: Autonomous AI Research Analyst

2025 – 2026 School year

- Built an autonomous Python pipeline to ingest arXiv cs.AI papers biweekly, applying a **three-layer filtering system** (keyword scoring, author boosting, Gemma 3 27B LLM evaluation) to surface the **top 25 most impactful** from ~1500 submissions per cycle, auto-deployed as GitHub Pages digests.
- Developed a low-latency RAG Telegram bot leveraging a **hybrid SQLite vector and keyword store** backed by **Gemini 2.5**, for real-time grounded querying of the curated papers in **<5 seconds**.

Skills

Languages & Frameworks: Python, SQL, TypeScript, FastAPI, Next.js, Node.js, Snowflake, PostgreSQL

ML, Infra & other: PyTorch, XGBoost, MLOps, Azure, AWS, Docker, Proxmox, Tailscale, Cloudflare, GitHub Actions, Linux, Git & Github for version control